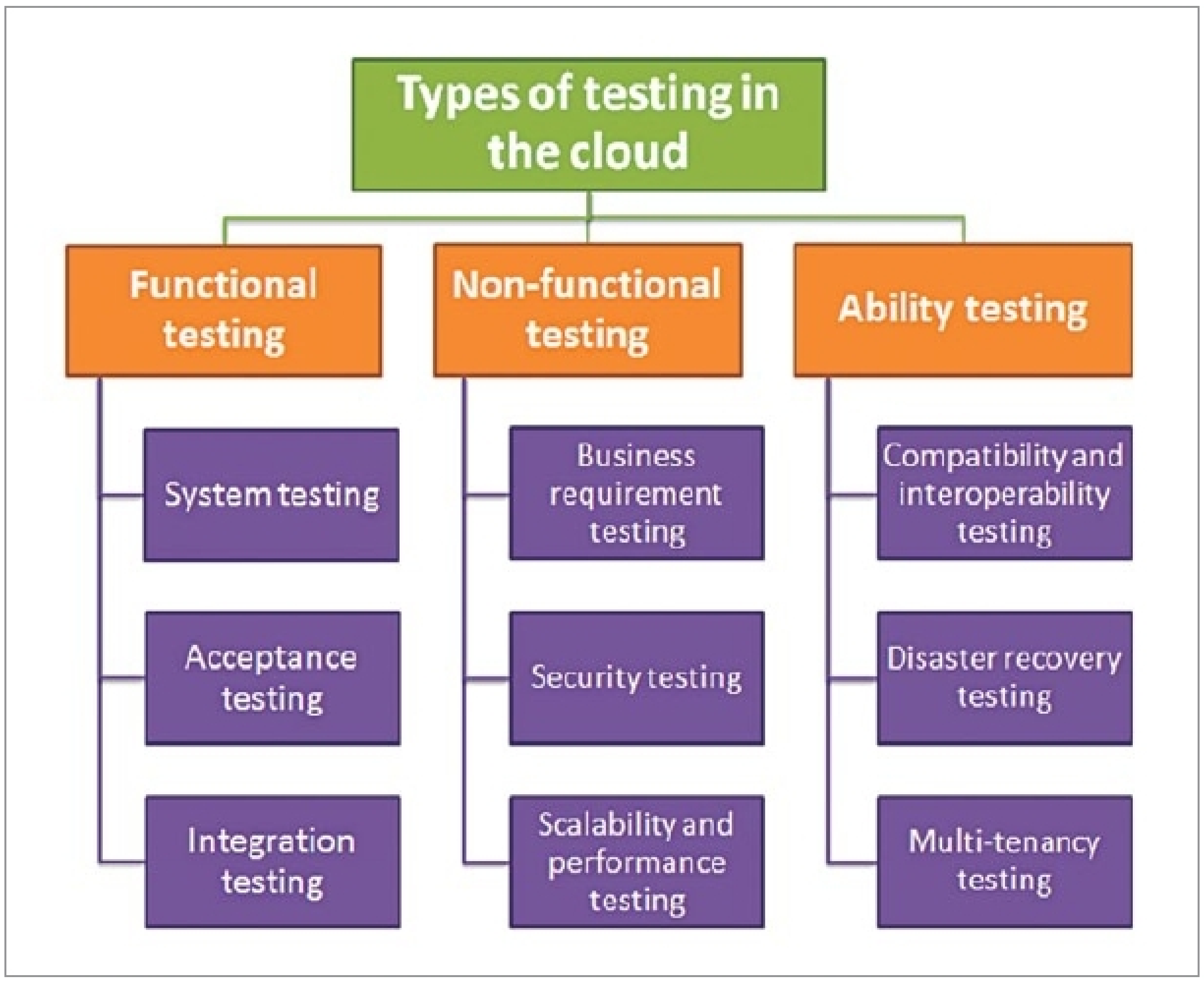
data is stored and transmitted safely.

- Scalability and performance testing to measure software response time while the system is subjected to increasing load.

***Ability testing.*** This verifies whether users really receive application services on demand. To check this, the test team can conduct the following types of testing:

- Compatibility and interoperability testing to evaluate the application's compatibility with various environments and platforms.
- Disaster recovery testing to evaluate disaster recovery time and ensure that the application becomes available to users again with minimum data loss.
- Multi-tenancy testing to verify whether the application can ensure a sufficient level of security and access control when multiple users invoke it in the cloud.




*Types of testing in the cloud (Credit: www.apriorit.com)*

## Challenges of cloud-based testing

Although cloud-based testing has many benefits for assuring quality, it also has operational challenges that the testers should be ready to overcome. Vendors of cloud-based tools provide terms and conditions of their cloud services that differentiate the responsibilities of the vendor and the cloud user.

There is a lack of standards for how testers can integrate internal resources of their companies' data centres with public cloud resources. Since architectures and operating models are developed by public cloud providers, these cloud services have little interoperability. Thus, testers may face compatibility issues if they want to switch cloud vendors because different cloud vendors might not offer the same services.

Security in the cloud still raises many concerns, as encryption techniques are far from perfect. There are still concerns about security of data that may be stored in a remote location beyond a company's legal jurisdiction. While providers guarantee round-the-clock availability of their services, even the least bit of downtime can cause negative consequences to testing processes.

Before selecting cloud-based testing tools, it is necessary to make sure the provider offers all configurations, storage and technologies that are needed, as it may be difficult to emulate customer environments if some configurations are not supported by the provider. Moreover, creating a test environment that includes all necessary settings and data can be time-consuming for testers.

Although, vendors inform their clients about prices for their cloud-based services, improper use of test environments may significantly increase costs. To avoid hidden costs, testers should thoroughly plan their test environments to take into account additional costs such as for data encryption and monitoring the use of cloud resources.

## Future of cloud technology in measurement

Cloud-based testing offers Web-based access to real devices spread globally and connected through live networks, providing end-to-end control for manual and automated testing practices. For teams spread across different locations, cloud-based test management platforms make collaboration possible. Users can log in from anywhere in the world, at any time, on any device and get access to the test environment.

Cloud-based tools provide real-time report generation and upgradation. This makes it infinitely easier to share important data, track and communicate with each other. A tester in any company office can connect to the cloud and select the machine he or she wants to test an application on. Every time the tester adds a piece of code, tests it and then redeploys it, the app immediately moves to production and release. This allows all members of the project team to collaborate in real time on a test so that problems can be identified and resolved rapidly.

Cloud-based tools have 24/7 support, and users should seek a contract where they are compensated for any downtime. Cloud-based tools reduce IT management tasks like installation, adding or replacing users, licensing and upgrades in systems across geographies. Users have a simple browser log-in that is accessible on any device, and they should be able to add or remove licences whenever they need to. Cloud-based test management tools are flexible and agile-friendly. EFY